

Curriculum Vitae

Megan Fairchild

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EDUCATION

Doctor of Philosophy in Mathematics

Louisiana State University, Baton Rouge, Louisiana

May 2025

Doctoral Thesis: Non-Orientable Surfaces Bounded by Knots

Master of Science in Mathematics

Louisiana State University, Baton Rouge, Louisiana

May 2022

Texas State University, San Marcos, Texas

December 2018

Thesis: Using Knot Theory to Model and Analyze DNA Replication and Recombination.

Bachelor of Science in Mathematics *Cum Laude*

Texas State University, San Marcos, Texas

May 2016

Minor: Computer Science

HONORS & AWARDS

FELLOW: Research Training Groups (RTG)

2023-2024 & 2024-2025

Pasquale Porcelli Graduate Student Academic Excellence Award

2021

Texas State Three Minute Thesis (3MT) People's Choice

2018

"Math in the picture" contest winner

2017 & 2018

Academic Excellence, Department of Mathematics

2017 - 2018

Academic Achievement, Department of Mathematics

2014 - 2016

Dean's List

2012 - 2016

TEACHING

Graduate Instructional Assistant - Louisiana State University

Fall 2020 - May 2025

INSTRUCTOR: Calculus I

LEAD TA: Business Calculus

Lecturer - Texas State University

Fall 2019 - Summer 2020

INSTRUCTOR: Contemporary Mathematics (*Quantitative Reasoning*)

INSTRUCTOR: Co-Requisite Intermediate and College Algebra

INSTRUCTOR: Co-Requisite Elementary Algebra and Contemporary Mathematics

Graduate Instructional Assistant - Texas State University

Spring 2017 - Fall 2018

INSTRUCTOR: Intermediate Algebra

TA: Calculus II, Pre-Calculus

Undergraduate Instructional Assistant

Fall 2014 & Fall 2015

TA: Pre-Calculus

Paper Grader

Spring 2016

Modern/Abstract Algebra, Introduction to Advanced Mathematics

Tutoring/ Math Tutoring Lab at Texas State

Fall 2014 - Fall 2017

Topology, Modern/Abstract Algebra, Differential Equations, Linear Algebra,

Calculus I, II, & III, Pre-Calculus, College Algebra

PUBLICATIONS & PREPRINTS

THE NON-ORIENTABLE 4 GENUS OF 11 CROSSING NON-ALTERNATING KNOTS. 2024
The Journal of Knot Theory and its Ramifications
doi: 10.1142/S0218216524500500

NON-ORIENTABLE 4 GENUS OF TORUS KNOTS 2025
to appear in The Journal of Knot Theory and its Ramifications
Preprint: arXiv:2405.04737

Submitted for Publication:

TOPOLOGICAL DATA ANALYSIS OF MORTALITY PATTERNS DURING THE COVID-19 PANDEMIC 2025
Preprint: arXiv:2510.15066

WHITEHEAD DOUBLES AND NON-ORIENTABLE SURFACES 2024
Preprint: arXiv:2410.13736

NON-ORIENTABLE SURFACES IN KNOT TRACES 2024
Preprint: PDF available upon request

Research Notes & Course Projects:

Brunnian Braids and Their Relation to Homotopy Groups of Spheres. 2022
Using Knot Theory to Model and Analyze DNA Replication and Recombination. 2018
The Betti Numbers of the Data Set of the Work of Bob Ross. 2018
Analyzing Betti Numbers of Range Image Patches. 2018
Analyzing Betti Numbers of the Zodiac Constellations. 2018
The History of the Alexander Horned Sphere. 2017
The History of i , the Imaginary Number. 2017
The Eccentricity of Directed Graphs. 2012

PROFESSIONAL SERVICE

SUPERVISOR OF UNDERGRADUATE RESEARCH:

Led vertically integrated research course Fall 2022-May 2025

Topological Data Analysis
(1, 1)-Knots and Heegaard Floer Homology
Non-Orientable 4 Genus of Torus Knots
Braids and Configuration Space

Oversaw research via directed reading Spring 2024-May 2025

Math Biology - The Tangle Model
Seifert Surfaces and the Knot Signature
Homological Algebra
3-Manifolds and Heegaard Diagrams
Introduction to Knot Floer Homology

CURRICULUM DEVELOPMENT:

Co-Coordinator of Contemporary Mathematics, Texas State University Spring 2020
Re-structured syllabus, created course material still being used

CREATOR & ORGANIZER:

The Math Empowerment Group - University of Utah Fall 2025-current

OFFICER:

Mathematics Student Colloquium Committee Fall 2022 - May 2025

ORGANIZER:

Informal Geometry & Topology Seminar, Louisiana State University	Summer 2024-May 2025
Grad Students in 4D, Louisiana State University	Spring 2023-May 2025

MENTAL HEALTH ALLY:

GEAUX Math LSU (new graduate student orientation)	Fall 2021-May 2025
On the list of 'people to talk to' since Fall 2021, panelist for surviving graduate school	
Certified Ally at Texas State University	Fall 2019

WEBSITE DEVELOPMENT:

General Exam Preparation	Summer 2024
Created a website to help younger graduate students prepare for their general exam	
Editor of the Faculty webpage for the University of Utah CDA	Fall 2025

ADMINISTRATION

The University of Utah	June 2025-current
<i>Associate Director - Center for Disability and Access</i>	
Faculty Outreach	
Data Analyst	
Supervisor: Deaf and Hard of Hearing Department	
Hiring Committee for the Health & Wellness Department	
Universal Design & Access Committee	
Instructor Accessibility Guidelines Group	
University Webmaster	
Staff Council Member	
New Staff Outreach Committee	

COLLOQUIA, CONFERENCE, & SEMINAR TALKS*Invited Talks:*

Joint Mathematics Meetings at Seattle, WA	2025
<i>Non-Orientable Surfaces in Knot Traces.</i>	
AMS Sectional Meeting at the University of Kansas	2025
<i>Non-Orientable Surfaces in Knot Traces.</i>	
University of Nevada Reno Math Colloquium	2024
<i>The Non-Orientable 4 Genus of 11-Crossing Non-Alternating Knots.</i>	
Texas State University Topology Seminar	2024
<i>Non-Orientable Surfaces Bounded by Knots.</i>	
AMS Sectional at University at Albany	2024
<i>Non-Orientable Surfaces Bounded by Knots & the Knot Trace.</i>	
Topology Students Workshop at Vanderbilt	2024
<i>The Non-Orientable 4 Genus of Knots.</i>	
Topology Conference at University of Georgia	2024
<i>Non-Orientable Surfaces in Knot Traces.</i>	
AMS Sectional at Florida State University	2024
<i>The Non-Orientable 4 Genus of Knots.</i>	
Geometry & Topology Seminar (LSU)	2024
<i>The Non-Orientable 4 Genus of 11-Crossing Non-Alternating Knots.</i>	
Kansas State Student's Topology Seminar	2024
<i>Whitehead Doubles & Non-Orientable Surfaces.</i>	

<i>Informal Topology Seminar & Other Research Presentations:</i>	
Slicing Obstructions from 4-Manifold Theory	2024
Algebraic Knot Concordance Group	2024
The Double Branched Cover and its Relation to Non-Orientable 4 Genus.	2024
The Non-Orientable 4 Genus of Torus Knots.	2024
The Non-Orientable 4 Genus of Knots.	2023
Smooth versus Topological Concordance via Whitehead Doubles.	2023
Brunnian Braids and Their Relation to Homotopy Groups of Spheres.	2022
Legendrian Knots & DGA.	2021
Modeling DNA Replication and Recombination with Knot Theory.	2018
Analyzing Betti Numbers of Range Image Packages.	2018
The History of Imaginary and Complex Numbers.	2017
The History of the Alexander Horned Sphere.	2017
Heroes of Homology.	2017

CONFERENCES ATTENDED

AMS Sectional - University of Kansas	March 2025
Nebraska Conference for Undergraduate Women in Mathematics	January 2025
Joint Mathematics Meetings - Seattle, WA	January 2025
AMS Sectional - University at Albany	October 2024
ICERM Braids Reunion Workshop - Brown University	July 2024
Topology Students Workshop - Vanderbilt University	June 2024
Topology Summer School & Conference - University of Georgia	May 2024
AMS Sectional - Florida State University	March 2024
Texas Geometry and Topology - Rice University	November 2023
SEC Topology - University of Alabama	June 2023
Topology Conference - University of Georgia	May 2023

LANGUAGES

Spanish	<i>Intermediate Speaker</i>
Coding: C++, JavaScript, Python, MatLab, HTML, L ^A T _E X	

MEMBERSHIPS

American Mathematical Society	May 2025
Math Club, Texas State University	2012- 2018
Pi Mu Epsilon, Texas State University	2016 - 2018
National Society of Collegiate Scholars	2013 - 2018